IEOR E4004: Optimization Models and Methods
Your Name:

In Class Exercise

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Consider the linear program:

$$
\begin{aligned}
\textbf{(P)} \quad \max \quad & 4x_1 + 3x_2 \\
\text{s.t.} \quad & x_1 + 2x_2 \leq 8 \\
& 3x_1 + 0.5x_2 \leq 9 \\
& x_1 - x_2 \geq 1 \\
& x_1 + x_2 \leq 6 \\
& x_1, x_2 \geq 0
\end{aligned}
$$

Question:

Without solving the LP directly, try to find a number U such that $4x_1 + 3x_2 \leq U$ for every feasible solution.

Hint:

- Multiply each constraint by a number of your choice.
- Add the inequalities together.
- Choose multipliers carefully so that the left-hand side becomes greater than or equal to $4x_1 + 3x_2$.
- Your multipliers must respect the inequality directions.